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Engineering Portfolio

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> I work at the intersection of **computational physics and control**. My research resolves turbulence and acoustic source fields at HPC scale for the US Army Corps of Engineers; my design work turns those methods into hardware, from federally deployed underwater deterrent systems to hand-operated brick presses. This portfolio documents seventeen projects across CFD, FEA, controls, and machine learning — each stated as the problem encountered, the method chosen and why, and the outcome achieved.

$340K	**76M**	**13**	**25%**
Federally funded program led (US Army ERDC × Virginia Tech)	Peak cell count in verified CFD grid-convergence study	Graduate capstone teams supervised as ME 569 TA	Peak structural mass reduction at constant stiffness

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FLAGSHIP RESEARCH — US ARMY CORPS OF ENGINEERS

PROJECT 01

Underwater Acoustic Deterrent System (“BASIC”) for Invasive Carp Control

Project Lead · $340,000 award · UW CAMP-PhyRE Lab × US Army ERDC × Virginia Tech | PI: Dr. John Palmore | 2024 – present

PROBLEM

Invasive silver and bighead carp are advancing up the Mississippi River system toward the Great Lakes. Navigation locks are the choke points where they can be stopped, and acoustic deterrents are the least ecologically invasive barrier available. But a lock chamber is the worst possible acoustic environment to predict: hard concrete walls, a pressure-release free surface, entrained air, and a strongly sheared through-flow. Free-field propagation models do not apply, and — critically — **the flow through the lock generates its own broadband noise**. If that self-noise overlaps the deterrent band, the barrier is masked by the very structure it protects. No prior study had quantified that flow-noise source field, so transducer placement and drive frequency were being set empirically.

METHOD

- **Stage 1 — Resolve the hydrodynamics.** Built a 3D RANS model of the lock approach, dogleg contraction and chamber (12 ft / 3.65 m depth, ~100 m streamwise). Solved in **COMSOL Multiphysics** and independently in **OpenFOAM** so that no conclusion rests on a single code.
- **Stage 2 — Prove the grid is trustworthy before trusting the physics.** Ran an **11-case grid convergence study** from 4M to 76M cells, sweeping inflation-layer count (1 → 30+) and first-layer thickness (5 mm → 0.2 mm) to place y^+ deliberately across the wall-function and resolved-sublayer regimes. Benchmarked every case against the universal law of the wall (Spalding), Coles' law of the wake, and the velocity-defect law, and compared **k-ε against k-ω SST on identical meshes** to isolate closure error from discretization error.
- **Stage 3 — Reconstruct the acoustic source.** Applied a **Stochastic Noise Generation and Radiation (SNGR)** method to synthesize a divergence-free turbulent velocity field from the steady RANS k and ε fields, then evaluated the **Lighthill turbulence source tensor** T_{ij} from it — recovering unsteady acoustic source information from a steady solve at a fraction of the cost of LES/DNS.
- **Stage 4 — Map it.** Sampled **10,000 probe points** through the chamber, computed source spectra at each, and assembled spatial maps of broadband source level to identify where the lock is acoustically loudest.
- Executed the full campaign on an **HPC cluster under SLURM** with multigrid solvers (AMG in COMSOL, GAMG in OpenFOAM) and domain decomposition.

IMPACT

Delivered the operational **BASIC** deterrent system to US Army ERDC, replacing empirical siting with a physics-based specification. The source-field maps show broadband Lighthill levels of **139–143 dB re 1 µPa** concentrated near the dogleg shear layer, and a source spectrum peaking at **1.22 Hz** — well below the carp hearing band. That separation is the design-enabling result: it demonstrates that deterrent signals can be placed in a frequency window where the lock's own hydrodynamic noise does not mask them, and identifies the wall regions to avoid when siting transducers.

$340K
Federal award led across three institutions

11 grids · 76M cells
Formal convergence study, two independent codes

10,000 probes
Source points sampled for spectral mapping

Deployed
Operational system delivered to ERDC

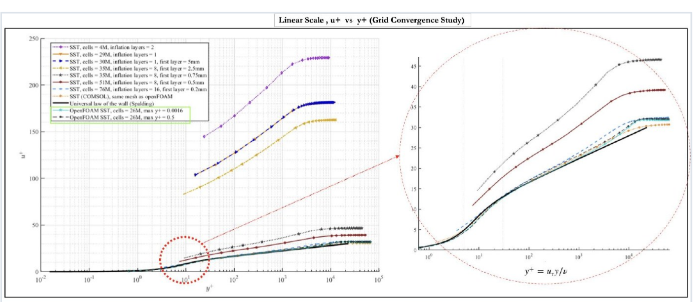


Fig. 1.1 — Grid convergence: u^+ vs y^+ across 11 meshes (4M–76M cells, SST and k-ε, COMSOL and OpenFOAM) against the universal law of the wall. The inset shows the converged band; outlying curves are under-resolved wall treatments, not physics.

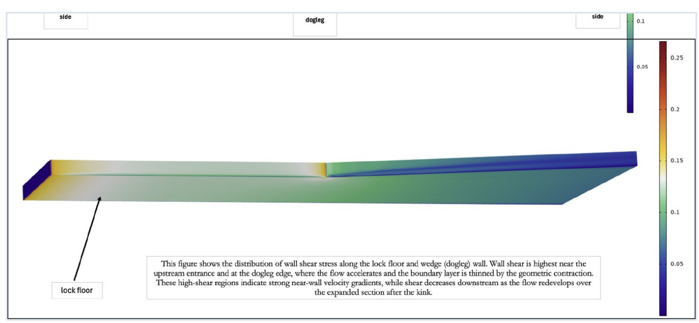


Fig. 1.2 — Tangential wall shear on the lock floor and dogleg wall. Shear peaks at the geometric contraction where the boundary layer thins, then relaxes as the flow redevelops — the same region that dominates acoustic source strength.

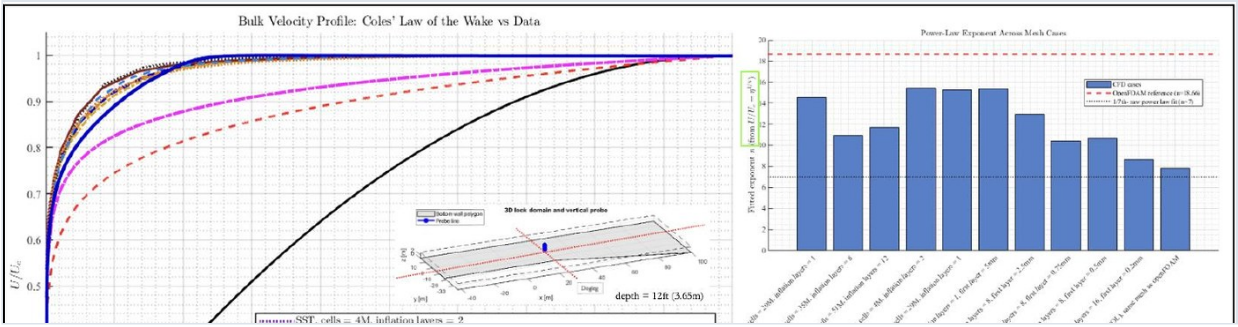


Fig. 1.3 — Bulk velocity profiles against Coles' law of the wake, with the fitted power-law exponent extracted for every mesh case. Cross-code agreement between COMSOL (AMG) and OpenFOAM (GAMG) on matched meshes is the verification evidence underpinning all downstream acoustic results.

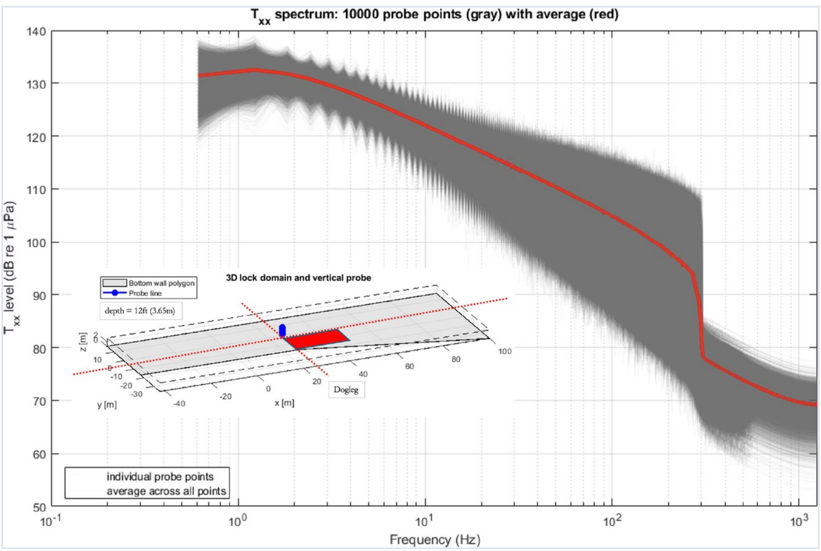


Fig. 1.4 — Lighthill source tensor T_{xx} level spectrum at all 10,000 probe points (grey) with the ensemble average (red). Broadband roll-off is captured to the synthesis cutoff; the ensemble peak sits at 1.22 Hz.

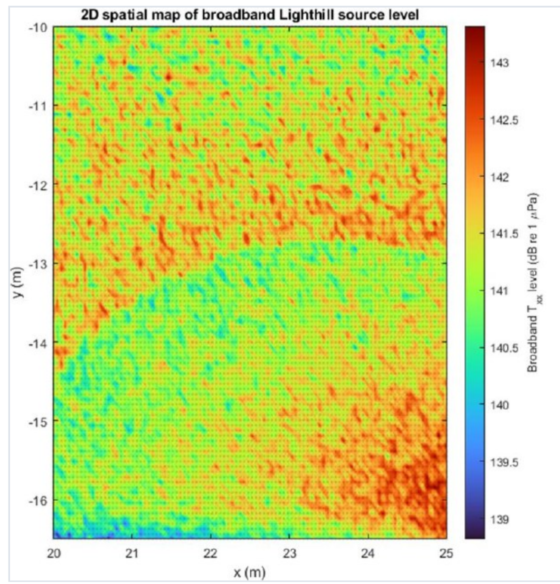


Fig. 1.5 — Spatial map of broadband source level (dB re 1 μPa) over a 5 m × 6 m patch downstream of the dogleg. Hot bands mark shear-layer structures; these are the regions to avoid when siting deterrent transducers.

TOOLS COMSOL Multiphysics 6.3 · OpenFOAM · MATLAB · Python · SLURM / HPC · CUDA

PROJECT 02

Turbulent Kinetic Energy Budget Closure in Multiphase Sprays

Research Assistant · CAMP-PhyRE Lab, University of Washington | PI: Dr. John Palmore

PROBLEM

Engineering multiphase spray models close the turbulence budget with isotropic eddy-viscosity assumptions and a gradient-diffusion hypothesis for turbulent transport. Near the liquid core and in bubble-laden regions both assumptions fail: the Reynolds stress tensor is strongly anisotropic and transport is not down-gradient. The literature offers no term-by-term resolved budget against which to calibrate, so modellers tune constants blind.

METHOD

- Resolved the **full anisotropic Reynolds stress tensor** from high-fidelity multiphase fields rather than collapsing it to a scalar turbulent kinetic energy.
- Computed **third-order velocity moments** to evaluate the turbulent-transport term directly — the term conventionally absorbed into a model constant — closing production, transport, pressure–strain and dissipation independently.
- Quantified interfacial energy exchange to isolate where dispersed-phase feedback, rather than mean shear, dominates production.
- Implemented the analysis as a reproducible post-processing pipeline over time-resolved cluster-stored fields.

IMPACT

Produced term-by-term budget data addressing a documented gap in multiphase spray modelling, giving RANS and LES closures a calibration target instead of a tuned constant. This is the physical foundation on which the ERDC deterrent work (Project 01) rests.

TOOLS OpenFOAM · Python (NumPy/SciPy) · MATLAB · HPC post-processing

PROJECT 03

Teaching & Supervision — ME 569, Data-Driven Control & Machine Learning

Teaching Assistant · University of Washington | Supervisor: Dr. Steven L. Brunton | 2024 – present

PROBLEM

Graduate students arriving in a data-driven control course can implement algorithms but frequently cannot judge when a method is appropriate, whether an identified model is physically meaningful, or how to design a validation that would actually falsify their result.

METHOD

Ran office hours, method reviews and design critiques, then supervised and evaluated **13 graduate capstone teams** spanning Koopman-operator MPC, control-barrier-function reinforcement learning, sparse sensor placement with autoencoders, SINDy residual learning, tube MPC with LTV-LQR ancillary feedback, neural-network MPC for bioprocess optimisation, and LLM-augmented control pipelines. Assessment focused on whether the validation design could have detected failure — not on whether the plots looked good.

IMPACT

Built breadth across essentially the full modern data-driven control toolbox and the judgement to review other engineers' modelling work — the reviewing skill that transfers directly to industrial R&D. Several supervised projects reached conference-report quality.

DOMAINS REVIEWED Koopman operators · SINDy · CBF-QP safety filters · MPC variants · Reinforcement learning · Sparse sensing · Neural surrogates

PROJECT 04

Tractor-Towed Smallholder Crop Harvester

Lead mechanical designer · Appropriate-technology programme, Eastern Cape, South Africa | Concept → CAD → prototype documentation

PROBLEM

Smallholder farms in South Africa's Eastern Cape typically work 1–5 hectare plots. Commercial combines are economically impossible at that scale — they cost more than the land and cannot manoeuvre in small irregular fields — so harvesting is done by hand. Manual harvesting is the labour bottleneck of the season: it sets a hard ceiling on plantable area, and because it is slow, a large fraction of the crop is lost to over-ripening and weather before it can be lifted. The engineering requirement was a machine that a cooperative of several farms could afford collectively, tow behind a common utility tractor, and repair with local welding and hardware.

METHOD

- **Architecture chosen for repairability, not performance.** A ground-engaging lifting deck feeds an inclined belt-and-cleat elevator that discharges into a trailing collection bin. Every drive element is a flat belt or chain on plain bearings — no hydrostatic drives, no proprietary gearboxes, nothing that cannot be replaced from a rural hardware supplier.
- **Elevator geometry set from first principles.** Cleat pitch and belt inclination were sized against the static friction angle of the harvested material so that product is carried rather than rolling back, with belt speed matched to forward ground speed to avoid the two failure modes at either extreme: bunching at the deck throat, and product throw at the discharge.
- **Frame sized by FEA under field loading.** The welded box-section chassis was analysed in SolidWorks under combined towing draft, deck reaction and a shock case representing a struck obstruction, checking von Mises stress at weld joints and deflection at the elevator mounts. Section sizes were chosen from locally stocked stock sizes, then verified — rather than optimised to a size nobody sells.
- **Single-point PTO drive** from the tractor, with a shear-pin coupling as the deliberate sacrificial element so that a jam destroys a cheap replaceable pin instead of the elevator drivetrain.
- Full assembly modelled in SolidWorks with a fabrication drawing package dimensioned for manual (non-CNC) fabrication.

IMPACT

Design-stage modelling projects a roughly **6–8× reduction in labour-hours per hectare** against hand harvesting, and — more consequentially — compresses the harvest window enough to cut weather- and over-ripening losses substantially. At an estimated build cost around **one-twentieth of the smallest commercial combine**, the machine is within reach of a multi-farm cooperative, which is the ownership model the design assumes. The wider intent is to remove the labour ceiling that currently caps how much land a smallholder family can plant.

Fig. 4.1 — Full assembly render. The inclined cleated elevator discharges over the rear of the towing frame into a collection bin; the ground-engaging deck (foreground) is height-adjustable at the hitch to suit row profile. Box-section frame members and plain-bearing shafts were selected for field repairability.

TOOLS SolidWorks · FEA (static structural) · Hand calculation of belt/cleat kinematics · Design for manual fabrication

Quantified outcomes for Projects 04 and 05 are design-stage engineering projections from modelling and cost estimation, not measured field results.

PROJECT 05

Manual Hydraulic Brick Press for Community Construction

Lead mechanical designer · Appropriate-technology programme, Eastern Cape, South Africa | Concept → CAD → prototype documentation

PROBLEM

Housing and school construction in rural South African communities is constrained less by labour than by the delivered cost of materials. Fired bricks are manufactured centrally and trucked in; over any real distance the transport cost exceeds the brick cost, and fired brick production is itself carbon- and fuel-intensive. Compressed stabilised earth blocks (CSEB) solve this — they are made from the soil already on site with roughly 5–10% cement — but the presses that make them either need three-phase power that rural sites do not have, or are imported machines priced out of community reach. The requirement was a press needing **no electricity at all**, buildable and maintainable by a village workshop.

METHOD

- **Human-powered hydraulic force multiplication.** A long hand lever drives a small-bore pump cylinder feeding a large-bore ram. Compaction pressure is set by the ratio of lever arm to piston areas, so the design problem reduces to choosing that chain of ratios such that one operator of ordinary strength reaches CSEB compaction pressure (order 2–4 MPa on the block face) at a sustainable pumping effort — the alternative, a direct mechanical toggle press, cannot reach that pressure by hand at usable block sizes.
- **Lever kinematics sized around the operator, not the machine.** Lever length and pivot placement were set from a comfortable standing push force and stroke, then checked so that peak effort occurs where the operator has best mechanical advantage rather than at full extension.
- **Mould box and ram FEA.** The mould walls and top platen were analysed under full compaction pressure for von Mises stress and, critically, for **wall deflection** — a mould that bows under load produces barrel-shaped blocks that will not course properly, so deflection rather than yield was the governing criterion.
- **Ejection designed as part of the cycle.** The same lever, re-indexed, drives block ejection through the top of the mould, so a full cycle needs one operator and no separate mechanism, and the hopper is positioned to allow refill during ejection.
- Frame welded from standard angle and box section; the only bought-in precision parts are the hydraulic cylinders, seals and a hand valve — all of which are automotive-jack-class components available regionally.

IMPACT

Because the machine converts on-site soil into a structural block, it removes brick transport from the build budget entirely — the dominant cost line on remote sites — with modelling suggesting a **materials cost reduction of roughly 60–70%** against delivered fired brick, and a large embodied-carbon saving from eliminating both firing and haulage. Requiring no grid connection, it is deployable at the site itself, and its economic significance is as much about **where the money goes** as how much: block production shifts from a distant factory to the community doing the building.

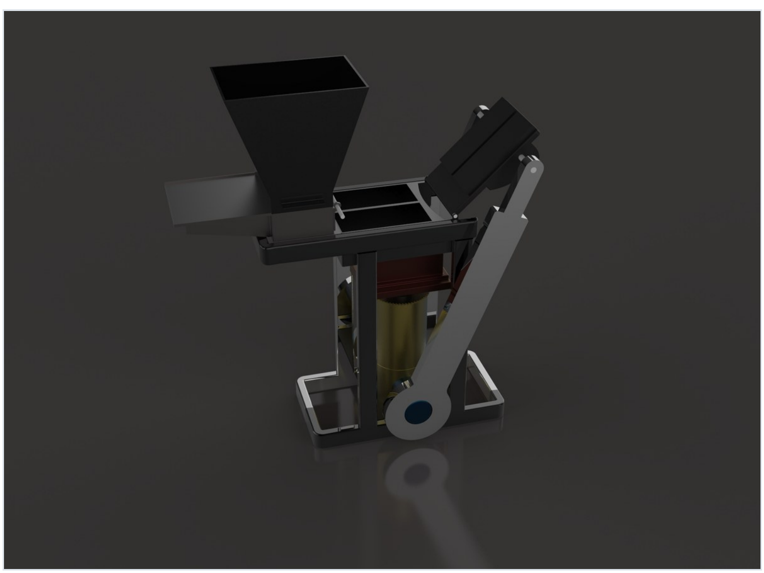


Fig. 5.1 — Press assembly render. Soil hopper (upper left) charges the mould box; the hand lever (right) drives the pump cylinder that pressurises the main ram visible within the frame. The hinged top platen swings clear for ejection. Frame is welded standard section throughout.

TOOLS SolidWorks · FEA (static structural, deflection-governed) · Hydraulic circuit sizing · Human-factors load analysis

PROJECT 06

Model-Predictive Attitude Control for Post-Capture Spacecraft

CAMP-PhyRE Lab · ME 569 Data-Driven Control & Machine Learning

PROBLEM

After capturing irregular orbital debris, a servicing spacecraft's inertia tensor changes abruptly and unpredictably and the combined body tumbles. Fixed-gain attitude control degrades under that plant change, and a naive controller commands torques the reaction wheels cannot deliver — saturation then destabilises the very recovery manoeuvre it was meant to execute.

METHOD

- Formulated rigid-body attitude dynamics — quaternion kinematics with Euler's equations — and linearised about the operating point for a receding-horizon optimisation.
- **Chose MPC specifically because saturation is a hard constraint, not a penalty.** Actuator limits and slew rates enter as inequality constraints on the QP; an LQR can only discourage them through weighting and will still command what it cannot deliver.
- Included disturbance terms for residual debris dynamics; tuned horizon length and state/input weights against the settling-time versus control-effort trade-off; solved the constrained QP each step.
- Validated across three deliberately distinct regimes — **large pointing-error slew**, **pure de-tumbling**, and **combined displacement plus rate** recovery — chosen because each stresses a different part of the constraint set.

IMPACT

Re-acquired target attitude in all three scenarios with no actuator-limit violations, demonstrating a constraint-aware alternative to gain scheduling for on-orbit servicing and active debris removal.

TOOLS MATLAB · Python · Constrained QP solvers · Quaternion attitude dynamics

PROJECT 07

SINDy Reduced-Order Flight Dynamics inside a Real-Time MPC Loop

CAMP-PhyRE Lab · JSBSim 6-DoF simulation | Data-driven system identification

PROBLEM

High-agility flight leaves the linear envelope where tabulated aerodynamic derivatives are valid, but a full 6-DoF nonlinear model is far too expensive to evaluate inside an onboard optimiser. Guidance is therefore forced to choose between agility and real-time feasibility. A black-box neural surrogate would solve the cost problem while creating a worse one: a flight-dynamics engineer cannot inspect it, and it cannot be certified.

METHOD

- Generated aggressive-manoeuvre datasets in **JSBSim (6-DoF)** with excitation designed to persistently excite coupled longitudinal–lateral modes — without that, sparse regression recovers only the modes the pilot happened to fly.
- Applied **SINDy** over a polynomial and trigonometric candidate library with sparsity-promoting regression, recovering interpretable state-space equations rather than an opaque map.
- Verified that surviving terms correspond to physically meaningful **aerodynamic cross-couplings** (roll–yaw coupling, angle-of-attack dependence) rather than fitting artefacts, and cross-validated on held-out manoeuvres outside the training envelope.
- Embedded the sparse model as the prediction model in an **MPC** trajectory-tracking loop, so each horizon evaluation costs a handful of algebraic terms instead of a full-fidelity integration.

IMPACT

Real-time-capable high-agility trajectory tracking with a model compact enough for onboard guidance and transparent enough to be reviewed term by term — a practical route toward certifiable data-driven flight control.

TOOLS JSBSim · Python · SINDy / sparse regression · MPC · Reduced-order modelling

PROJECT 08

RL–MPC Hybrid Control and Sim-to-Real Pipeline for a FANUC CRX Collaborative Arm

CAMP-PhyRE Lab · Reinforcement learning with optimal-control safety layer

PROBLEM

Reinforcement learning explores well but offers no constraint guarantee on real hardware; MPC guarantees constraint satisfaction but depends on a model that never matches the physical arm. Policies trained purely in simulation also collapse on deployment through the reality gap. Industrial deployment tolerates none of these failure modes: an unguarded learned policy near a human is not acceptable regardless of its average performance.

METHOD

- Built a **hybrid architecture** in which the RL policy proposes task-level actions and an MPC layer enforces joint limits, velocity bounds and workspace constraints before any command reaches the controller — performance from learning, feasibility from optimisation.
- Trained in simulation with **domain randomisation** over inertial parameters, joint friction and control latency, so the policy is forced to be robust to exactly the quantities that are misidentified in practice.
- Designed the **sim-to-real transfer pipeline**: matched state and action interfaces, control-rate alignment, and staged deployment from simulator to hardware.
- Instrumented tracking error and constraint-violation rate to compare pure-RL, pure-MPC and hybrid baselines on the same task.

IMPACT

Demonstrated learned manipulation behaviour that remains constraint-feasible on real collaborative hardware, and produced a transfer pipeline reusable across tasks on the same platform.

TOOLS Python · PyTorch · MPC · Domain randomisation · ROS · FANUC CRX

PROJECT 09

Distributed Multi-Agent Coordination with Control-Barrier-Function Safety

Supervised student teams · Crazyflie-class quadrotors and ground robots

PROBLEM

Heterogeneous aerial and ground robot teams must hold formation and avoid obstacles and each other using only local information. Centralised planning does not scale with team size, and a learned policy alone cannot certify collision avoidance — it can only be shown not to have collided yet.

METHOD

- Implemented **distributed formation control** using neighbour-only communication so computational and communication cost stay bounded as the team grows.
- Layered a **control barrier function quadratic program (CBF-QP)** as a minimally invasive safety filter: it modifies the nominal command only when the forward-invariance condition on the safe set would otherwise be violated, so performance is untouched except at the safety boundary.
- Trained **RL policies** for nominal navigation and let the CBF-QP certify their output — combining learned performance with a formal guarantee rather than choosing between them.
- Validated in simulation and on hardware with Crazyflie-class quadrotors and ground robots in cluttered indoor environments.

IMPACT

Collision-free formation keeping and navigation on real hardware. The safety-filter architecture was adopted as a standard pattern across several subsequent student projects.

TOOLS Python · CBF-QP · Multi-agent RL · Crazyflie · Convex optimisation

PROJECT 10

Remaining Useful Life Prediction for Engine Prognostics & Health Management

Center of Advanced Research in Engineering & AI, NUST | Submitted manuscript | NASA C-MAPSS dataset

PROBLEM

Turbofan maintenance scheduled on fixed intervals either replaces healthy components early or misses degrading ones. Data-driven remaining-useful-life (RUL) prediction promises condition-based maintenance instead, but deep models for RUL are notoriously sensitive to hyperparameters — particularly the **sliding-window length**, which sets how much degradation history the network can actually see. Hand-tuned networks in the literature report good average error while degrading badly on the engines that matter most: those near end of life.

METHOD

- Built a **convolutional network** over multivariate sensor time series from the NASA C-MAPSS dataset — two convolution + ReLU stages (64 then 32 filters, kernel sizes 3/4/2), pooling, flattening and fully connected regression to a scalar RUL. Convolution is the right inductive bias here: degradation signatures are local temporal patterns that recur at unknown offsets.
- Wrapped training in a **multi-objective optimisation** framework tuning window length, kernel size, batch size, epochs and steps-per-epoch together, because these interact — window length and kernel size jointly determine the receptive field in cycles, so tuning them independently finds a false optimum.
- Ran a data-visualisation stage first, tracking sensor channels (e.g. HPC outlet temperature) against RUL to confirm which channels carry genuine degradation signal before committing them to the model.
- Evaluated per-engine predicted-versus-actual RUL trajectories rather than aggregate error alone, so late-life divergence could not hide inside a good mean score.

IMPACT

Optimisation markedly tightened prediction against ground truth, with the largest gains in the final cycles before failure — precisely the region that determines whether a maintenance decision is actionable. Work submitted as a manuscript from the Center of Advanced Research in Engineering & AI.

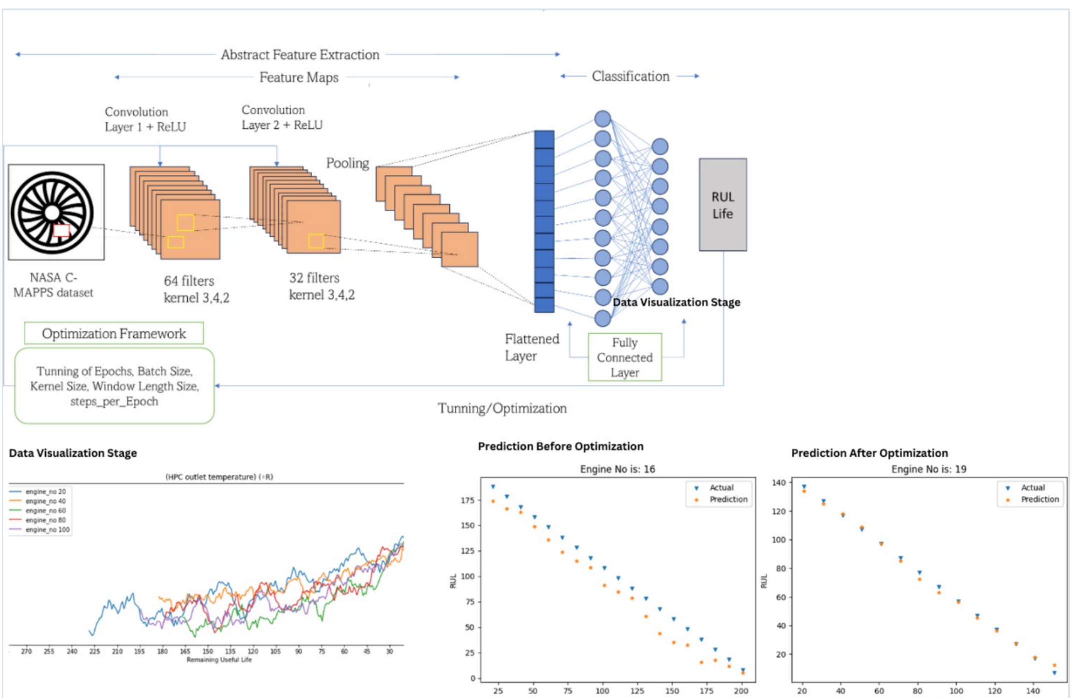


Fig. 10.1 — CNN architecture and optimisation framework (top), with sensor-degradation visualisation and predicted-versus-actual RUL trajectories before and after hyperparameter optimisation (bottom). Post-optimisation prediction tracks ground truth through end of life rather than diverging late.

TOOLS Python · TensorFlow/Keras · NASA C-MAPSS · Multi-objective optimisation · Time-series feature analysis

PROJECT 11

Aircraft Weapons Bay Design Optimization

National Aerospace Science and Technology Park (NASTP) | Cavity aeroacoustics & store separation

PROBLEM An open weapons bay in flight is an aerodynamic cavity: the shear layer spanning the opening drives intense resonant pressure oscillations that fatigue structure, shake avionics, and create an unsteady flow field that makes store release unpredictable. The existing bay used straight internal collars, which do little to disrupt that resonance.

METHOD

- Established solver credibility first on the **lid-driven cavity benchmark** — a 4:1 aspect-ratio case with published horizontal and vertical velocity-component profiles — before trusting any result on the real geometry. Recirculation structure and profile shape were matched against reference data.
- Modelled the bay in **ANSYS Fluent**, extracting static pressure fields and velocity streamlines to locate the primary recirculation and the shear-layer impingement point on the aft wall, which is where cavity resonance is driven.
- Ran a **parametric comparison of collar geometries** — straight (baseline) against **backward-leaned** and **yawed** collars — on the reasoning that leaning and yawing the collar thickens and de-correlates the shear layer, breaking the coherent feedback loop that sustains resonance rather than merely deflecting flow.
- Compared candidate designs on recirculation strength and aft-wall pressure loading.

IMPACT Produced a benchmarked optimised bay configuration with weakened cavity recirculation and reduced aft-wall pressure loading relative to the straight-collar baseline, supporting quieter carriage and more predictable store separation.

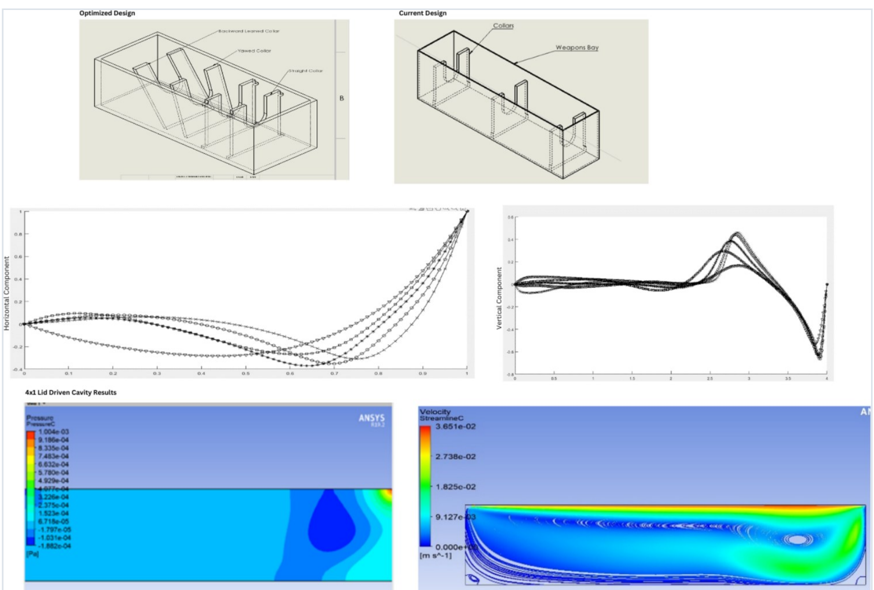


Fig. 11.1 — Optimised versus baseline bay geometry with backward-leaned and yawed collars (top); lid-driven cavity validation profiles (middle); Fluent static-pressure and velocity-streamline fields showing the aft recirculation core (bottom).

TOOLS ANSYS Fluent · Cavity-flow benchmarking · Parametric geometry study

PROJECT 12

Aerodynamic Optimization of a Formula Student Racing Vehicle Shell

Formula Electric Club × High Performance Computing Research Lab

PROBLEM The existing bodyshell measured **C_d = 0.5**. On an energy-limited electric Formula Student car, drag is a direct tax on range and top speed, and the target set for the redesign was **C_d = 0.32** — a 36% reduction that could not be reached by smoothing surfaces alone. The shell also had to wrap an existing tubular spaceframe whose hardpoints were fixed.

METHOD

- Built the shell as a **lofted surface through a series of parallel section planes** referenced to the spaceframe, so body sections became explicit design variables and the packaging constraint was satisfied by construction rather than checked afterwards.
- Ran external aerodynamic simulation in **ANSYS Fluent** on HPC resources, extracting static pressure fields and decomposing the force into **pressure and viscous components separately** — the decisive diagnostic, since pressure-dominated drag calls for changing the rear closure while viscous-dominated drag calls for reducing wetted area, and treating them as one number leads to reshaping the wrong end of the car.
- Iterated the aft body taper against the stagnation and separation behaviour visible in the surface pressure distribution, working the section profiles rather than local fairings.

IMPACT

Established a parametric, section-driven aero workflow that ties every shape change back to a measurable force component and can be re-run for each season's chassis, replacing intuition-led body styling with a quantified drag-reduction path toward the C_d = 0.32 target.

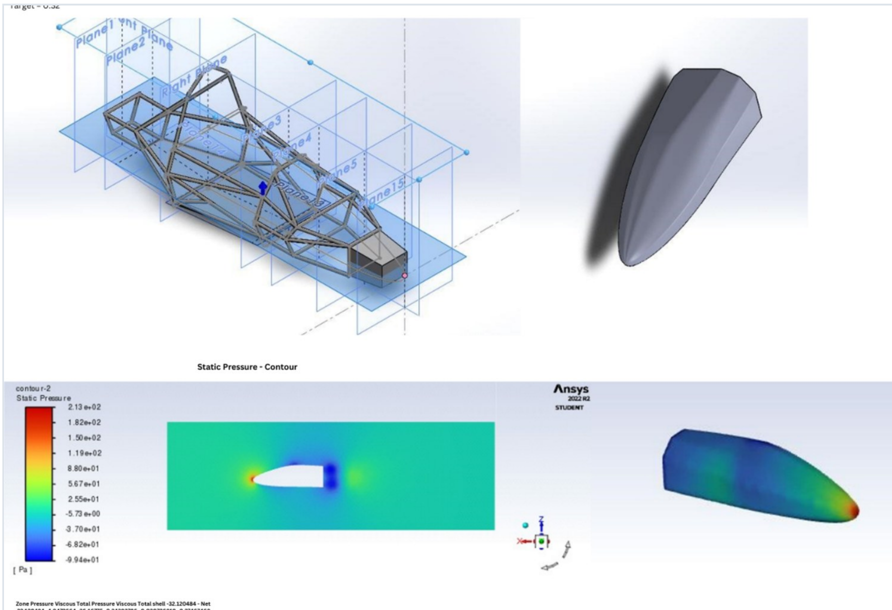


Fig. 12.1 — Section-plane construction over the spaceframe and resulting lofted shell (top); static-pressure contour through the centreline and surface pressure distribution with the stagnation region at the nose (bottom).

TOOLS ANSYS Fluent · SolidWorks surfacing · HPC · Force decomposition

PROJECT 13

Vertical-Axis Wind Turbine Sliding-Mesh Simulation — “Wind Genie”

Startup concept & ongoing research | Harvesting highway truck-wake energy

PROBLEM

Heavy highway traffic continuously injects energy into the air beside the roadway, and that wake energy is simply dissipated. Capturing it needs a turbine that works in **rapidly reversing, highly turbulent, direction-agnostic flow** — conditions in which a horizontal-axis turbine, which must yaw to face the wind, is useless. Simulating one is the harder problem: the rotor moves relative to the surrounding domain, so a single static mesh cannot represent it.

METHOD

- Selected an **H-type vertical-axis rotor**, which accepts flow from any direction without yawing — the governing requirement for a roadside installation where flow reverses with every passing vehicle.
- Split the computational domain into **stationary and rotating zones coupled by an Arbitrary Mesh Interface (AMI)**, allowing the rotor mesh to physically rotate while conserving flux across the sliding interface — the standard rigorous treatment for cross-flow rotors, and far more faithful than a frozen-rotor approximation for a machine whose whole physics is azimuthal blade loading variation.
- Ran unsteady RAS in **OpenFOAM (pimpleFoam)** with the k-ω closure and a Newtonian transport model: $\omega = 50$ rpm (5.23 rad/s), $\nu = 1.94 \times 10^{-5}$ m²/s, $\Delta t = 0.1$ s, ~150,000 cells, hierarchical domain decomposition at 12 tasks per node.
- Extracted blade-level aerodynamic loading through the rotation cycle and complemented it with wind-tunnel measurement of blade-airfoil lift and drag coefficients.

IMPACT

Produced resolved azimuthal blade-loading data identifying optimal turbine siting relative to the traffic lane, and underpinned a B2C concept placing roof-mounted units on green buildings. This is the project where I built the moving-mesh and parallel-decomposition skills later applied to the ERDC lock work.

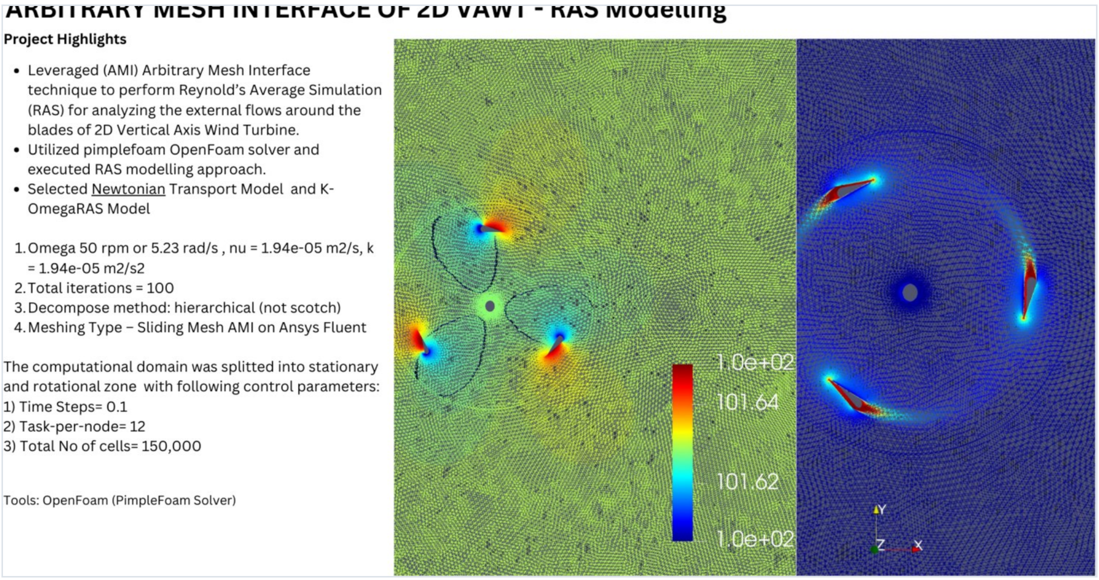


Fig. 13.1 — Rotating and stationary mesh zones coupled through the AMI, with pressure field around the three-blade H-rotor (left) and resolved blade wakes showing azimuthal loading variation (right).

TOOLS OpenFOAM (pimpleFoam, AMI) · ANSYS Fluent sliding mesh · Wind-tunnel testing · HPC

MECHANICAL DESIGN & STRUCTURAL ANALYSIS

PROJECT 14

Formula SAE Structural Program — Chassis, Suspension and Steering

Design & CFD Intern · Formula SAE, Virginia Tech | Aug – Dec 2023

14a · Topology-optimised chassis subcomponents

PROBLEM Lap time is mass-sensitive, but the accessible mass sits at welded chassis joints — exactly where stress concentrations and crash-structure requirements make naive material removal dangerous.

METHOD Ran **density-based topology optimisation** on critical weld-joint brackets with a compliance objective under a volume-fraction constraint and prescribed bump, braking and cornering load cases. Re-interpreted the optimiser output into **weldable, machinable geometry** rather than shipping raw organic shapes that the team could not fabricate, then re-verified the manufacturable interpretation with FEA (von Mises stress and deflection) to confirm the factor of safety survived the reinterpretation.

IMPACT **10% mass reduction** at critical weld joints with all stress constraints satisfied; the freed mass budget was redirected into the driver-protection structure for a **5% improvement in crash protection**.

14b · First-principles rear suspension

PROBLEM Suspension geometry carried over from previous cars was neither weight-optimal nor matched to the current mass distribution, and commercial kinematics packages hid the sensitivities the team needed in order to tune.

METHOD Wrote **custom MATLAB scripts** deriving suspension kinematics from first principles — A-arm hardpoint placement, roll-centre migration, camber and toe curves, and load transfer — so every design variable stayed inspectable and could be swept. Traded camber gain against roll-centre stability across the hardpoint space, propagated the resulting member loads into structural sizing, and validated engine mount, sprocket and powertrain brackets by FEA under those computed loads. Released manufacturable SolidWorks/NX models toleranced for in-house fabrication.

IMPACT **6% subsystem weight reduction** with validated structural margins, plus a reusable parametric toolchain letting the team re-evaluate geometry in minutes rather than re-modelling CAD.

14c · Lightweight steering wheel and mounting interface

PROBLEM The steering assembly sits high and far forward — poor for both mass and yaw inertia — yet any loss of torsional stiffness degrades steering feel, and the assembly must satisfy Formula Student packaging and ergonomic rules.

METHOD Traced the **load path** from grip to column and removed material only where it carried no steering torque, leaving the torsionally active structure intact. Fixed the ergonomic and rulebook packaging envelope as a constraint at the start rather than checking compliance afterwards. Validated by FEA under steering-torque loading against von Mises limits and an explicit deflection ceiling set to preserve feel.

IMPACT **25% structural mass reduction** with stiffness preserved under torque loading and full Formula Student packaging compliance.

−10%	−6%	−25%	+5%
Chassis joint mass	Suspension subsystem weight	Steering structural mass	Driver crash protection

TOOLS SolidWorks · Siemens NX · Topology optimisation · FEA · MATLAB · GD&T

PROJECT 15

Fixed-Wing VTOL and Autonomous Quadcopter UAV Program

Design lead · Hokie Flying Club, Virginia Tech | Concept → prototype → flight test

PROBLEM A VTOL aircraft must be aerodynamically efficient in cruise while carrying the structural and propulsion penalty of vertical flight, under hard weight and factor-of-safety limits. Compounding this, each CFD case was being set up by hand, so the team could evaluate only a fraction of the configurations it wanted to within a season.

METHOD

- Ran **CFD campaigns in ANSYS Fluent** across the configuration space — fuselage surfacing, wing–rotor interaction, transition attitudes — and cross-checked against **wind-tunnel** measurement rather than trusting simulation alone.
- **Templated the workflow**: standardised meshing, boundary conditions and post-processing so that a new configuration inherited a validated setup instead of being rebuilt. This was the change that cut setup time **15–20%** — a process fix, not a solver fix.
- Designed compact fuselage surfacing for integrated electronics, with battery-locking mechanisms specified using **GD&T** for repeatable assembly across 3D-printed parts.
- Built **LiDAR/TOF-equipped 3D-printed prototypes** for autonomy testing and verified structure by FEA against weight and FoS targets.

IMPACT Delivered flying VTOL and autonomous quadcopter prototypes end to end while cutting simulation turnaround **15–20%**, letting the team explore materially more of the design space within one season.

TOOLS ANSYS Fluent · Wind tunnel · SolidWorks / NX · FEA · GD&T · LiDAR & TOF sensing

PROJECT 16

Shell Eco-Marathon Urban Concept Vehicle — "Merlaon"

Hokie Electric Vehicle Team, Virginia Tech | Chassis, suspension and body

PROBLEM Shell Eco-Marathon scores energy consumed per distance in simulated urban driving, so every gram and every count of drag is scored directly — while the vehicle must still pass a full safety structure inspection with a driver inside. Efficiency and occupant protection pull in opposite directions on exactly the same structure.

METHOD

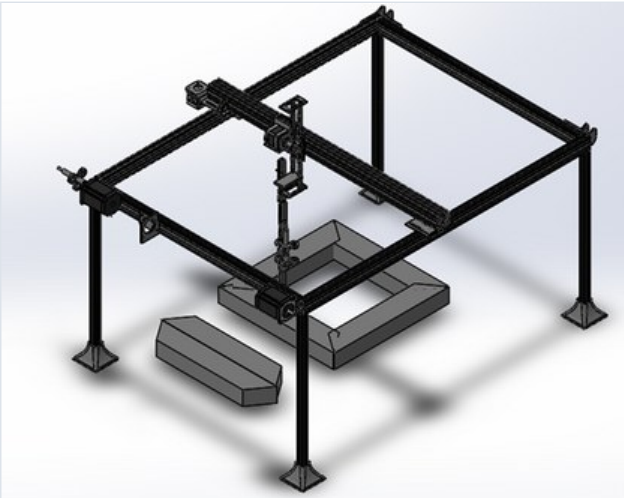
- Designed a **hybrid monocoque–spaceframe** architecture: a carbon-fibre central monocoque for torsional rigidity per unit mass, with TIG-welded 4130 chromoly front and rear subframes where load paths are concentrated and repairability matters. Used **FEA** to drive frame geometry to roughly **12 kg** while meeting roll-structure requirements, with integrated front crush zones for impact energy absorption.
- Adopted a **delta trike layout** (two front, one rear) to cut frontal area and rolling resistance, with enclosed wheel wells, smooth surfacing and front wheel fairings for a low-drag body.
- Specified a direct steering linkage at an **8:1 ratio** with **Ackermann compensation** to reduce tyre scrub in turns — scrub is parasitic energy loss, so steering geometry is an efficiency decision here, not only a handling one.
- Front double-wishbone suspension with adjustable coilovers and geometry chosen for minimal energy loss; centrally mounted electric motor for weight distribution and short power path; high-density lithium-ion pack with a BMS for safety and longevity.
- Precision-machined aluminium mounting points integrated into the chassis for suspension, powertrain and body-panel attachment.

IMPACT A competition vehicle in which each subsystem decision — layout, steering ratio, structural material split — traces to a quantified energy or safety requirement, achieving a ~12 kg frame without compromising the roll structure.

Fig. 16.1 — Full vehicle render showing enclosed wheel wells, low-profile body and integrated roll structure.

Fig. 16.2 — Spaceframe with triangulated bracing, driver packaging and delta-trike wheel layout.

TOOLS SolidWorks · FEA · CAE load-path optimisation · Composite & 4130 chromoly design



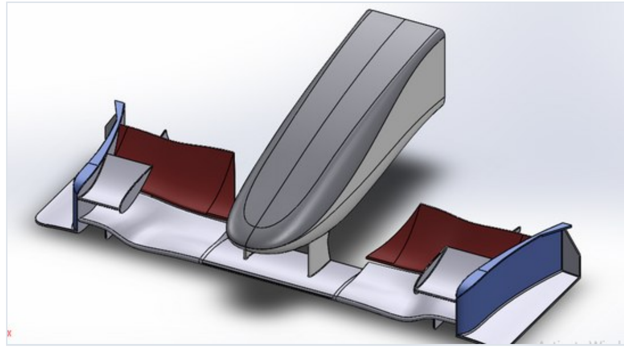
17a · 4-DOF Cartesian Robot for Transformer-Core Assembly

Concept → design → fabrication → deployment

Problem. Transformer cores are built by stacking thin silicon-steel laminations. Misalignment between sheets opens air gaps that raise reluctance and drive hysteresis and eddy-current losses, so stacking precision directly sets the efficiency of the finished transformer — and manual stacking cannot hold it consistently.

Method. Evaluated SCARA against Cartesian architectures and selected Cartesian: the workspace is a rectangular stack, and a Cartesian machine gives decoupled axes, position accuracy independent of pose, and far simpler calibration. Derived **inverse kinematics and path planning**, then optimised the trajectory for **minimum cycle time** under load, speed, acceleration and reliability requirements drawn from end-customer specification. Designed a vacuum end-effector for thin-sheet handling.

Conceptualised, designed, fabricated and implemented as a working line automation cell.



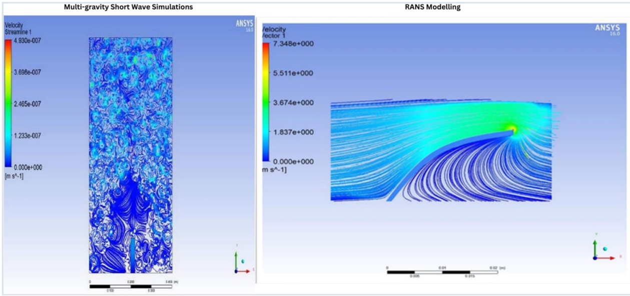
17b · Formula 1 Front Wing Aerodynamics

NUST F1 Student Team · CFD & surface modelling

Problem. The front wing is the first aerodynamic surface to meet clean air; it must generate downforce while managing the wake it sheds over everything downstream, since a poorly conditioned wake degrades the whole car's aero balance.

Method. Built the multi-element wing using **surface extrude and boundary-surface** features for control over section blending, produced ANSI-standard 2D drafting, and ran CFD focused on the first 10% of chord — where the suction peak forms and where separation, if it starts, originates.

Result: −4.139 kN lift (i.e. downforce) at 1.4 kN drag, quantifying the balance available for handling.



17c · Biomimetic Triboelectric Nanogenerator via Fluid–Structure Interaction

Fluid Dynamics & Bio-Inspired Design Lab · Submitted manuscript

Problem. Triboelectric nanogenerators convert mechanical deformation into charge, so in a flow environment their output is set entirely by how the flexible element deforms — which means the design problem is a coupled fluid–structure problem, not an electrical one.

Method. Ran **ANSYS FSI** studies coupling flow to a compliant biomimetic membrane: multi-gravity short-wave simulations resolving the turbulent free-surface field, and RANS modelling of flow over the deflected element to capture the separation and velocity concentration at its tip that drive oscillation.

Established the flow conditions under which the compliant element oscillates most energetically.

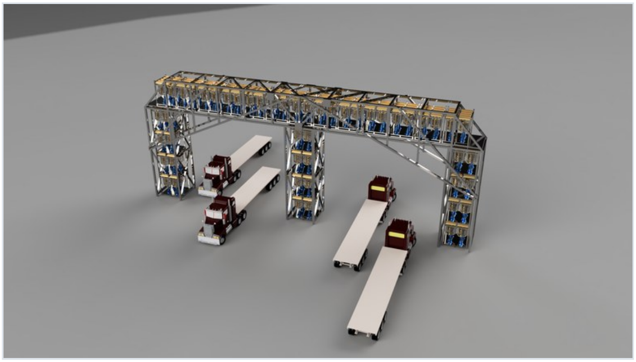
17d · "Open Flight Deck" Cockpit Concept

Human-factors-led industrial design · Fusion 360

Problem. Flight-deck workload is dominated by instrument scanning; poor situational awareness under high workload remains a persistent contributor to incidents, and conventional panels cannot be reconfigured as avionics evolve.

Method. Used **form modelling (T-spline) surfacing** in Fusion 360 to shape console geometry around reach and sightline envelopes rather than styling first and checking ergonomics after. Replaced fixed instruments with reconfigurable touchscreen surfaces so the deck can be updated in software and customised per operator.

A flight deck whose information layout is a software decision, improving situational awareness through automation and integrated avionics.



17e · Wind Genie — Highway Energy Harvesting Concept

Startup concept · System & business design

Problem. The wake energy shed by heavy highway traffic is continuously dissipated, while roadside communities and buildings draw grid power metres away.

Method. Integrated arrays of the vertical-axis rotors of Project 13 onto **truss structures spanning the carriageway**, siting them from the CFD-resolved wake field, and paired the infrastructure concept with a B2C roof-mounted product line for green buildings.

A deployment concept placing generation at the point of consumption, using flow that is otherwise entirely wasted.

17f · Automatic Scooping Brush

Freelance design engineering · Product & electronics

Problem. A fashion-industry client needed a hairbrush that delivered consistent vibration massage without the user having to supply the motion, in a housing indistinguishable from an ordinary brush.

Method. Packaged a **12 × 20 mm micro vibration DC motor** inside the brush body, switched by relay from an integrated power source, and validated the circuit in **Proteus** before committing to hardware. Made the head double-sided — vibrating bristles on one face, conventional bristles on the other — so a single product covers both use cases and the motor mass is balanced about the grip axis.

Simulation confirmed stable operation; delivered as a manufacturable product design with full electronics schematic.

PROJECT 18

Manufacturing Quality & Supply Chain Operations

Logistics Officer · Velo Group — British American Tobacco | Nov 2021 – Jan 2022 · Pakistan

PROBLEM

A new product line was ramping with unmeasured quality performance: defect and scrap rates were known only anecdotally, supplier consistency varied without consequence, and moulding losses were treated as a cost of doing business rather than a solvable engineering problem. A floor team of 32 reported through the role.

METHOD

- Instituted **key quality indicators** — defect rate, yield, equipment uptime — and validated the formulas, equipment and processes generating them, because a metric nobody trusts changes no behaviour.
- Ran a **Kaizen event** on moulding scrap with the floor team doing the root-cause work themselves, on the reasoning that the operators running the process daily already know where waste originates and need structure and authority rather than instruction.
- Built a **Supplier Quality Scorecard** giving real-time performance visibility, tied to quarterly reviews and formal corrective-action plans so measurement carried consequences.

IMPACT

35% reduction in scrap rate saving **$75,000 annually** in material cost on the finished-goods unit; **10% reduction in defects** through training and process optimisation; **20% improvement in supplier consistency**.

−35% Moulding scrap rate	**$75K/yr** Material cost saved	**−10%** Defect rate	**+20%** Supplier consistency

CAPABILITIES

Technical Skills

CFD & Multiphase	ANSYS Fluent · OpenFOAM (pimpleFoam, AMI/sliding mesh) · COMSOL 6.3 · STAR-CCM+ · RANS (k-ε, k-ω SST) · grid convergence and y^+ verification · Reynolds-stress & TKE budget analysis · FSI · cavity and free-surface flows
Acoustics	Finite-element acoustic propagation · Lighthill analogy & turbulence source tensors · Stochastic Noise Generation and Radiation (SNGR) · spectral analysis and PSD estimation · modal decomposition
FEA & Design	SolidWorks · Siemens NX · Fusion 360 (form/T-spline surfacing) · topology optimisation · von Mises and deflection-governed sizing · weldment and DFM design · GD&T · composite and steel structure
Control & ML	Model Predictive Control · LQR · Kalman filtering · SINDy & reduced-order modelling · Koopman methods · control barrier functions (CBF-QP) · reinforcement learning (DQN, policy gradient) · CNNs for time series · surrogate modelling · multi-objective and non-convex optimisation
Computing	Python (NumPy, SciPy, PyTorch, TensorFlow) · MATLAB & Simulink · C/C++ · CUDA · HPC with SLURM · Linux · Git · JSBSim · Proteus
Engineering practice	Verification & validation methodology · design of experiments · cross-code benchmarking · technical writing & peer review · cross-institution project leadership

EDUCATION & CONTACT

Background

PhD	Multiphase Flows & Acoustics — University of Washington, Seattle · expected June 2027
M.S.	Mechanical Engineering — University of Washington, Seattle · December 2025
B.E.	Mechanical Engineering — NUST, Pakistan · July 2020
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